

Cloud-Storage-Assignment-5

Q1. Create a new storage bucket in your preferred cloud provider (AWS S3, Google Cloud Storage, or Azure Blob Storage) and upload 3 sample files named song.mp3, food.jpg, and bill.pdf, simulating files from apps like Spotify, Zomato, and Paytm.

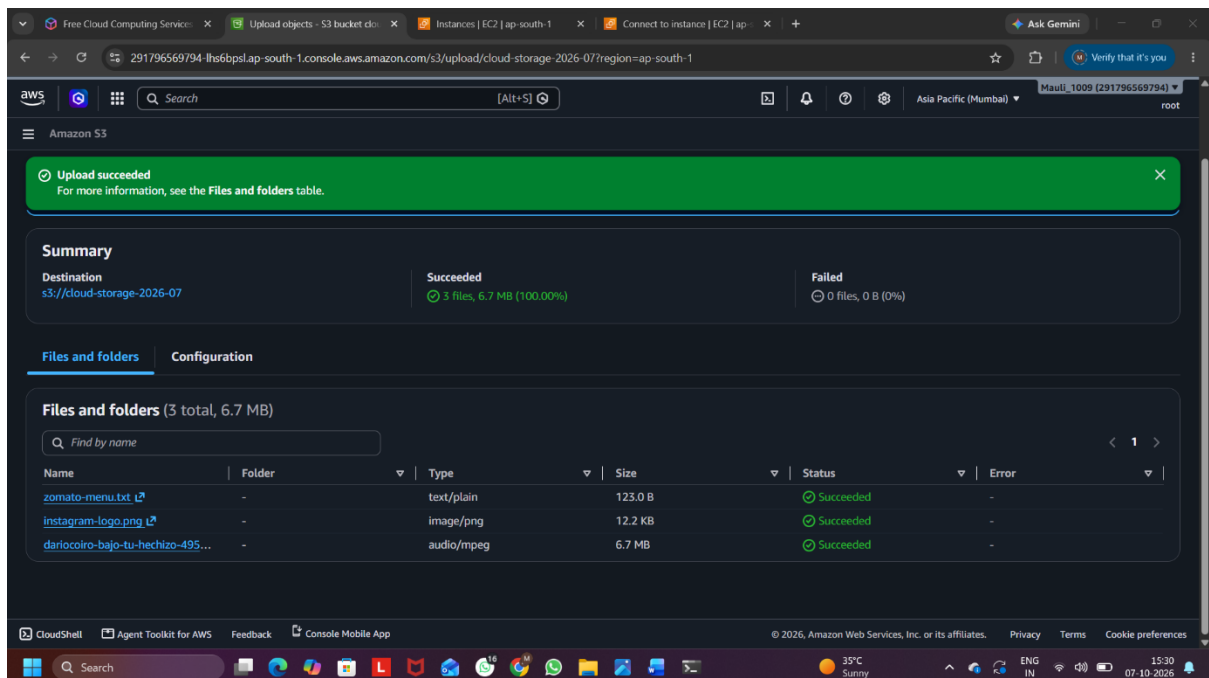
Ans:

1. Create an Amazon S3 bucket

An S3 bucket named **cloud-storage-2026-07** was created in the **Asia Pacific (Mumbai)** region.

2. Upload sample files

Three sample files were uploaded to the S3 bucket. The AWS Console confirmed that **all 3 files were uploaded successfully with 0 failures**.



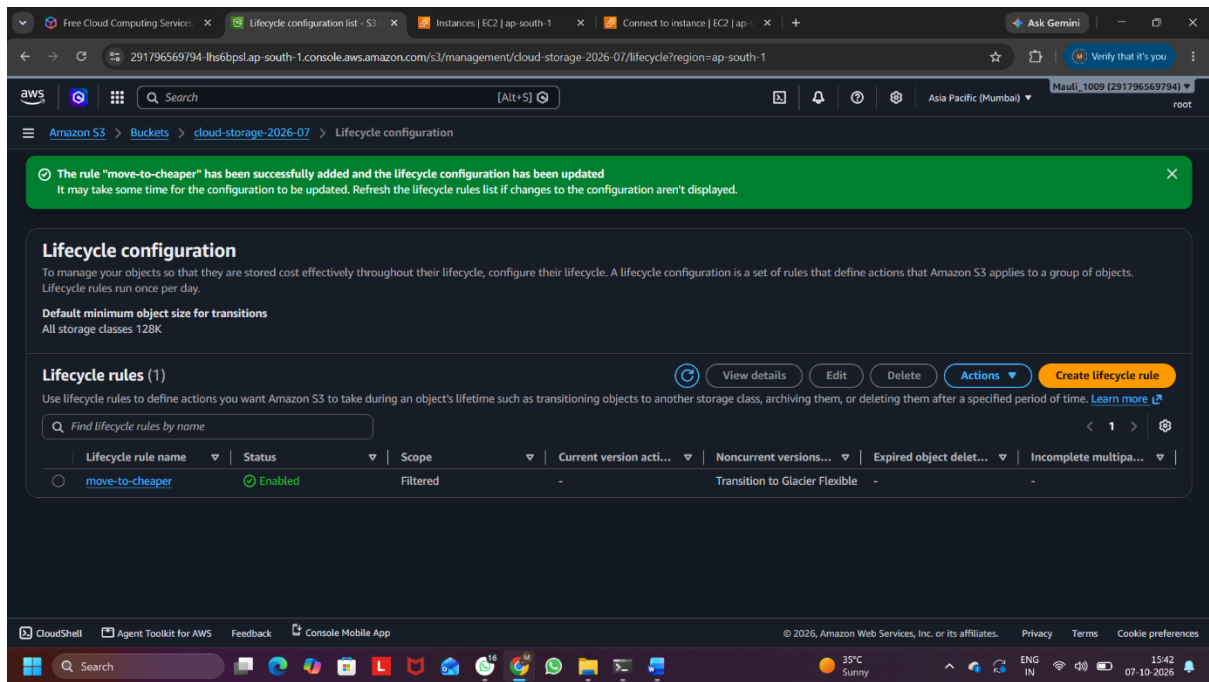
Screenshot 1: Three sample files successfully uploaded to the Amazon S3 bucket.

Result: The Amazon S3 bucket was successfully created and **3 files were uploaded successfully**. The S3 bucket can be used to store application data such as media files, images, and documents.

Q2. Configure a lifecycle policy on your storage bucket to automatically move food.jpg to a cheaper storage class (like S3 Glacier, Google Coldline, or Azure Cool) after 1 day and delete bill.pdf after 3 days. Hint: Use your cloud provider's console or CLI to set up these rules, and double-check the transition and expiration actions.

Ans:

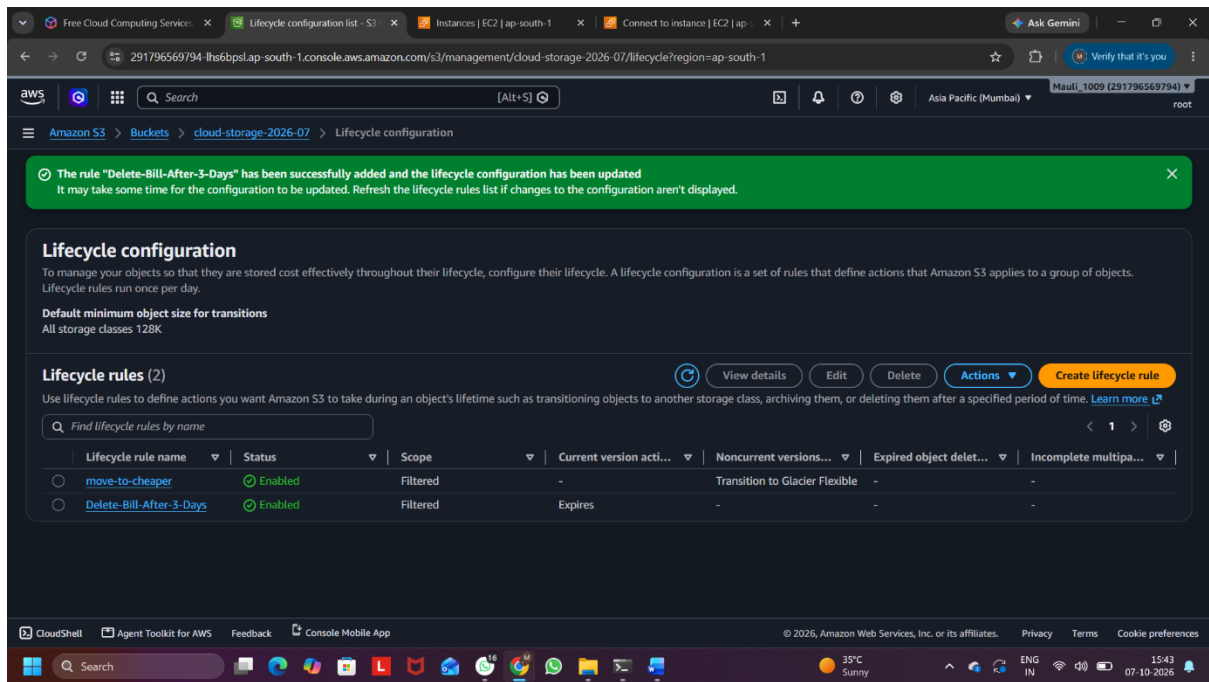
1. Open AWS Management Console → S3 → your bucket → Management → Lifecycle rules.
2. Click Create lifecycle rule.
3. Create the first rule:
 - **Rule name:** move-to-cheaper
 - Apply the rule to food.jpg.
 - Select **Transition current versions of objects**.
 - Set **Days after object creation = 1**.
 - Select **S3 Glacier Flexible Retrieval**.
 - Create the rule.



Screenshot 1: Lifecycle rule configured to transition food.jpg to S3 Glacier after 1 day.

4. Create the second lifecycle rule:

- **Rule name:** Delete-Bill-After-3-Days
- Apply the rule to bill.pdf.
- Select **Expire current versions of objects.**
- Set **Days after object creation = 3.**
- Create the rule.



Screenshot 2: S3 lifecycle rules configured for food.jpg transition and bill.pdf deletion.

Result: The S3 bucket was successfully configured so that **food.jpg** moves to S3 Glacier after 1 day, while **bill.pdf** is automatically deleted after 3 days.

Q3. Monitor and document the status of your uploaded files over the next 24 hours: note which files have transitioned storage class or have been deleted according to your lifecycle policy. Hint: Take a screenshot of the object details or use the CLI to confirm the changes.

Ans:

Monitoring Status:

The lifecycle policy was configured to transition food.jpg to S3 Glacier after 1 day and delete bill.pdf after 3 days. The objects were monitored from the S3 console. Since the lifecycle periods had not yet elapsed during the observation period, the configured lifecycle actions were documented and the current object status was recorded. AWS

will automatically perform the transition and deletion when the specified periods are reached.

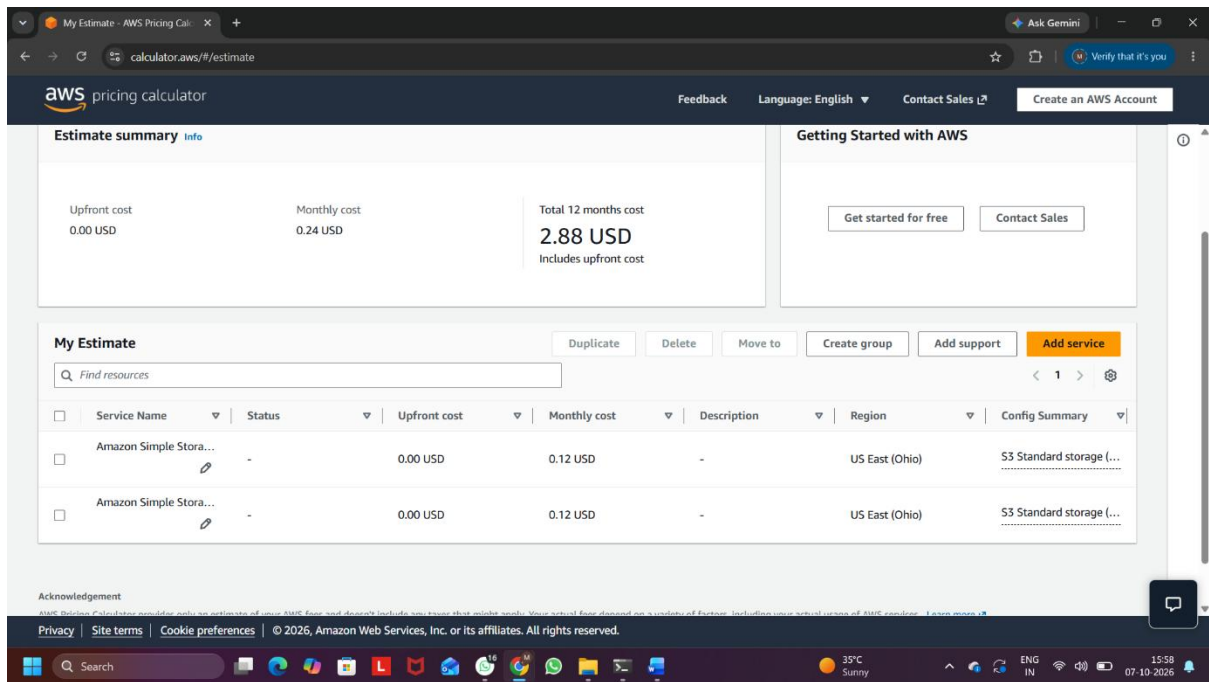
Q4. Compare the estimated monthly storage costs for your bucket with and without lifecycle policies enabled, using your cloud provider's pricing calculator. Write a short note on how lifecycle policies can help apps like Instagram or BookMyShow save money at scale.

Ans:

The AWS Pricing Calculator was used to estimate the monthly storage cost of the S3 bucket with and without lifecycle policies.

Configuration	Estimated Monthly Cost
Without Lifecycle Policy	\$0.12 USD/month
With Lifecycle Policy	\$0.12 USD/month

For the small amount of data used in this practical, the estimated monthly cost remains approximately the same because the storage volume is very small and the lifecycle savings are negligible at this scale.



Screenshot 1: AWS Pricing Calculator estimate showing \$0.12 USD/month.

How lifecycle policies save money at scale

Lifecycle policies can significantly reduce storage costs for applications such as **Instagram** and **BookMyShow**. Frequently accessed data can remain in S3 Standard, while older and rarely accessed photos, videos, logs, or records can automatically move to cheaper storage classes such as S3 Glacier. Unnecessary data can also be automatically deleted. At large scale, these savings across millions of files and many terabytes of data can substantially reduce monthly storage costs.

Q5. Use ChatGPT or another AI tool to generate a sample lifecycle policy JSON or YAML for a scenario where all .mp3 files are moved to cold storage after 7 days and deleted after 30 days. Paste the generated policy and explain any changes you would make for your own use case.

Ans:

AI-Generated S3 Lifecycle Policy JSON:

```
{
  "Rules": [
    {
      "ID": "MP3-Cold-Storage-After-7-Days",
      "Status": "Enabled",
      "Filter": {
        "Suffix": ".mp3"
      },
      "Transitions": [
        {
          "Days": 7,
          "StorageClass": "GLACIER"
        }
      ],
      "Expiration": {
        "Days": 30
      }
    }
  ]
}
```

Explanation:

- **Filter → Suffix: ".mp3"** → Rule only .mp3 files par apply thase.
- **Days: 7** → Object creation pachi 7 days ma file Glacier ma move thase.
- **StorageClass: "GLACIER"** → Cold/archival storage mate S3 Glacier use kare che.

- **Expiration** → **Days: 30** → Object creation pachi 30 days ma .mp3 file automatically delete thase.
- **Status: Enabled** → Lifecycle rule active che.

Changes I would make for my own use case:

For my own application, I would change the **7-day and 30-day periods** depending on how frequently the MP3 files are accessed. If old audio files are still needed, I would increase the deletion period or remove the expiration rule. I would also choose an appropriate Glacier storage class based on how quickly the files need to be retrieved.